# Bench 6 — ESP-IDF Filter Architecture Benchmark

Phase 6 comparative study over the clean ESP-IDF captures · 37 candidates · frozen configurations · corrected reconstruction

Hardware: 2× MS5611 GY-63 clones (foam-covered) + MPU6050, ESP32-S3 @ 240 MHz · firmware esp-idf/vario @ 8e79e68 · baro 40 Hz + 10 Hz temp, IMU 100 Hz

Status: complete frozen-configuration study with controls. **No estimator is selected or frozen.** Scripts: filter_benchmark/bench6/ (run6 → equiv6 → rate6 → tune6 → mcucpu6 → figures6); re-run on BMP581/LSM6DSV16X captures by registering the new stems in datasets.py.

# 1. Objective and datasets

Re-evaluate the full filter trade space on the clean ESP-IDF recordings — 29 standalone filters, stacked S1–S5, CV/Kalman references and IMU-fusion candidates — with the reconstruction defect DEF-001 corrected (fixed 101325 Pa ISA reference, firmware-exact compensation). This is a fresh comparative study, not a confirmation of previous rankings. Both barometers are evaluated independently throughout (the two clones differ; sid1 is consistently ≈1.15–1.4× noisier). Corrupted captures are excluded from scoring and retained as documented fault-test data.

***Table 1.*** *Scoring datasets and quality verdicts (measured, not labelled).*

| Capture | Role | Split | Quality |
|---|---|---|---|
| 221124_idf_elevator_2x | elevator run 1: 4 legs ±38.2 m, –1.72/+1.66 m/s | train | CLEAN |
| 221420_idf_elevator_2x | elevator run 2 (same profile) | validation | CLEAN (3× 20 ms gaps) |
| 215709_idf_handtest3 | static + slow moves + toss 148–152 s | train | GOOD (7× 20 ms gaps) |
| 234521_idf_p0fix_static | static 75.3 s | validation | CLEAN |
| 234944_idf_bbuf_check | static 20.2 s | validation | CLEAN |
| 221109_idf_elevator_2x | aborted capture | excluded | USB-stall dropouts (DEF-008) |
| 215027_idf_foam_static | foam static | excluded | CORRUPTED: 3703 seq gaps / 6192 dup IMU ts |

Controls: frozen configurations first; tuning is a separate leakage-controlled pass (train-select, validation-report); a sampling-rate control precedes all cross-chain comparisons; MCU-vs-replay equivalence precedes all replay-vs-firmware comparisons. Estimator output and virtual-audio behaviour are scored separately.

# 2. MCU equivalence and new firmware-chain findings

**Formal equivalence (queued task): PASS for D01.** On the post-p0fix statics, with the era-correct chain (fixed 101325 Pa reference against the firmware's 10× pressure scale — DEF-011 — plus the boot d2 seed), the Python replay reproduces MCU tone velocities with steady-state bias ≤ 0.3 mm/s, p95 |error| = 1.0 mm/s and 100 % audio ON/OFF transition agreement (24 and 37 transitions, both sids). C03 agrees to p95 71–105 mm/s (its gain-cache quantization). S4/S5/FUS emit no steady audio transitions on statics — a vacuous, honestly reported result.

**DEF-011 (new, major):** baro_proc.c::compensate_pa returns p01mbar·10, ten times the common.py scale. Under the moving-p0 firmware the ratio cancelled; the p0fix hardcoded 101325 Pa (common-line units) against the still-10× pressure, so post-fix firmware velocities are scaled by exactly 10^0.1903 = 1.5498 and absolute altitude is garbage (–23 725 m). Self-consistent within the firmware, so bench captures remain valid; equivalence replays must use the isa_fw10 chain.

**DEF-010 mechanism:** the boot d2 seed (prom[5]·256, 20 °C) steps the compensation at the first real T frame → startup audio flap (tones clamp at +6.67 m/s), amplified to a –1.5×10■ m/s transient by the DEF-011 scale; emulated by seed_d2=True, and startup (t < 8 s) is always reported separately.

**Gate-decision chaos (pre-p0fix moving window only):** during the first 50 s of the elevator/handtest captures, S4/S5 (and mildly C03/FUS) replay tones diverge from MCU by seconds, while the plain Kalman D01 — the core of both stacks — matches to ~1 mm/s everywhere. The raw reconstruction is

therefore right; the divergence lives in discontinuous gate latches amplifying sub-LSB micro-differences while the defective moving reference was active. It affects only old-firmware equivalence bookkeeping, never scoring (fixed-ISA), and vanishes with the frozen reference (frozen-era p95: D01 1.0–1.2 mm/s, S4 1.1–2.1 mm/s, S5 1.0–1.3 mm/s, all captures).

**Asymmetric tone clamp (new observation):** implied v = (f–500)/450 with f ∈ [400, 3500] Hz represents v ∈ [–0.22, +6.67] m/s; the firmware suppresses tones below 400 Hz, so MCU tone streams are blind to strong sink. Replay audio metrics are the authoritative comparison for sink behaviour.

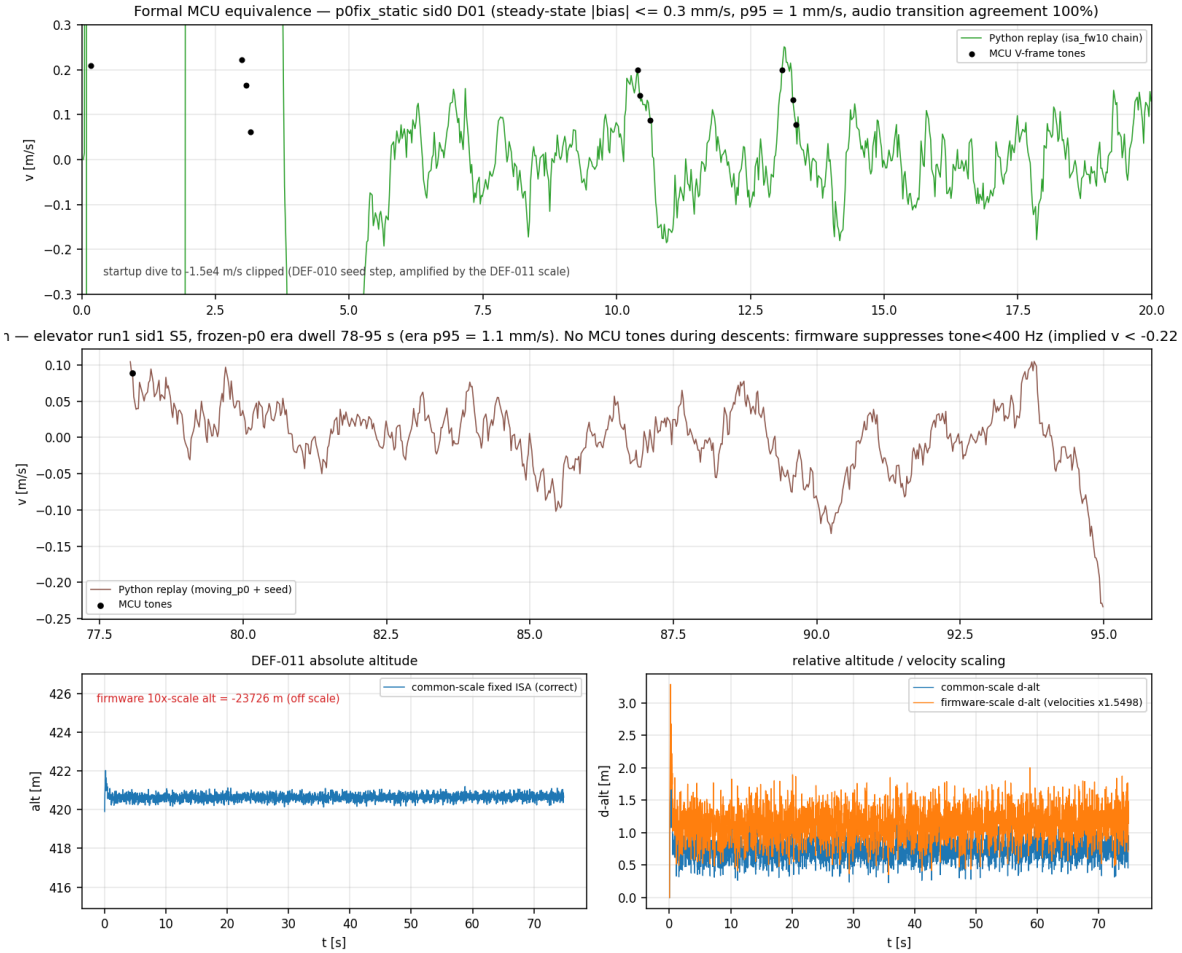



***Figure 1.*** *Formal MCU equivalence (a), harness validation in the frozen-p0 era (b), and the DEF-011 pressure-scale defect (c).*

# 3. Frozen-configuration results

## 3.1 Static silence

***Table 2.*** *Rest velocity RMS (m/s, medians). Phase 3/4 numbers were measured on the defective moving/session-mean reference — most of the apparent movement is the DEF-001 correction, not the filters.*

| Candidate | Phase 3/4 rest RMS | Bench 6 rest RMS | Δ |
|---|---|---|---|
| B02_project_kalman | 0.096 | **0.0165** | –83 % |
| C02_AlphaBetaGamma | 0.021 | 0.0217 | — |
| C01_AlphaBeta | 0.026 | 0.0282 | — |
| S1_RobustAdaptiveKF | 0.044 | 0.0346 | –21 % |
| S5_LightGateKF | 0.045 [M] | 0.0369 | –18 % |
| S4_FastSlowGated | — | 0.0371 | — |
| S2_DualTimescale | 0.084 | 0.0373 | –56 % |
| C03_CV_tracker | 0.048 | 0.0485 | — |
| D01_Kalman_CV | 0.101 | 0.0567 | –44 % |
| B02_Butter2 | 2.55 | 1.49 (rate artefact, §4) | — |

Audio false-latch on statics: 0 latches/h for the whole bank; the known chatterers remain A03_Holt (~5615/h), S3 (~4450/h), FUS_Q10 (~2273/h), E06 (111/h), D04 (55/h); unadaptive baselines B00/E01–E03 chatter at 6400–6700/h.

## 3.2 Elevator controlled motion (run 1, medians over sids)

***Table 3.*** *Sustained climb and descent track equally (\|bias\| ≤ 6 mm/s on ±1.7 m/s legs). The exit proxy is floored at ≈6.8 s by dwell wobble [D]. No filter missed any lift — elevator discrimination is saturated.*

| Candidate | leg RMS | leg bias | rise [D] | exit [D] | dwell false s | missed |
|---|---|---|---|---|---|---|
| C03_CV_tracker | 0.089 | –0.006 | 1.27 | 6.53 | 2.18 | 0 |
| S2_DualTimescale | 0.091 | –0.005 | 1.34 | 4.44 | 2.01 | 0 |
| S4_FastSlowGated | 0.091 | –0.006 | 1.34 | 6.07 | 2.15 | 0 |
| E06_RobustAdapt | 0.092 | –0.005 | 1.18 | 6.80 | 2.33 | 0 |
| D01_Kalman_CV | 0.093 | –0.005 | 1.26 | 6.80 | 2.27 | 0 |
| S1_RobustAdaptiveKF | 0.097 | –0.005 | 1.30 | 5.18 | 1.97 | 0 |
| S5_LightGateKF | 0.098 | –0.005 | 1.31 | 6.28 | 2.13 | 0 |
| C01_AlphaBeta | 0.106 | –0.004 | 1.67 | 3.30 | 2.21 | 0 |
| FUS | 0.241 | –0.028 | 1.77 | 6.81 | 4.95 | 0 |
| B02_Butter2 | 1.489 | –0.006 | n/m | 6.91 | 19.9 | 0 |

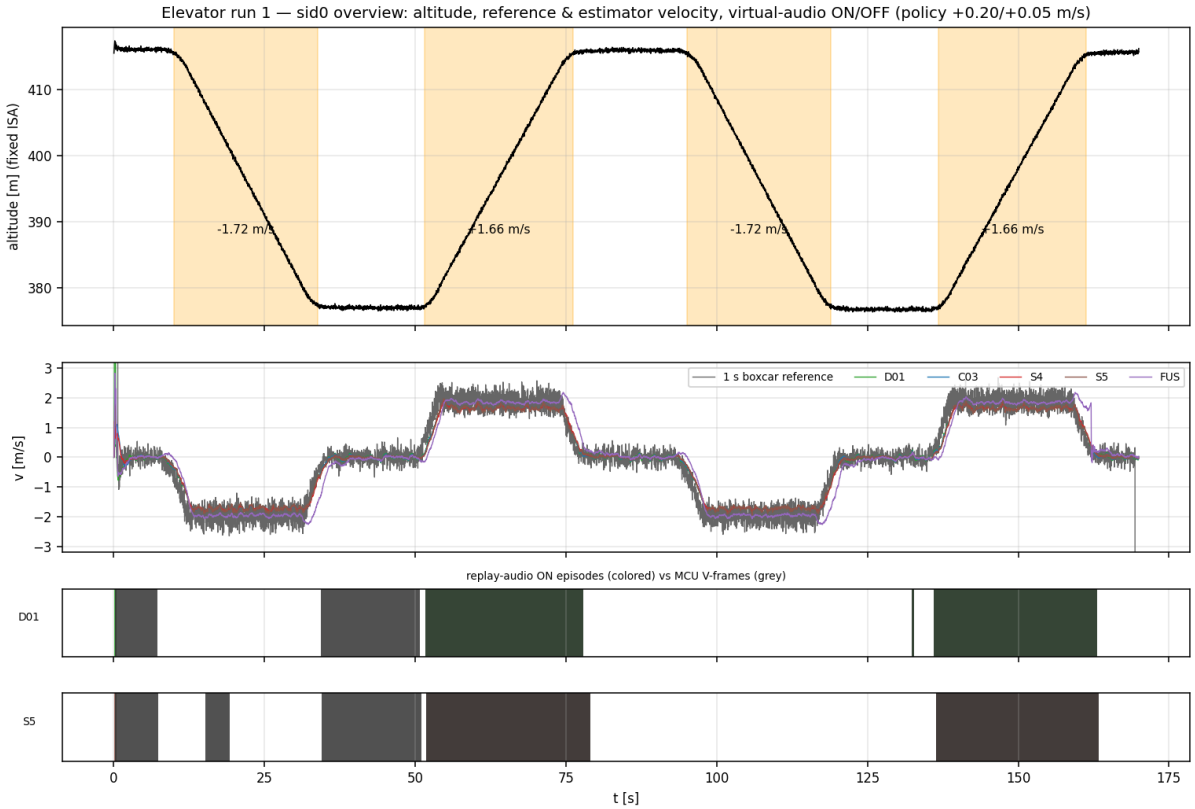



***Figure 2.*** *Run 1 overview: altitude with detected legs, bank velocity traces against the 1-s boxcar reference, and replay-audio ON episodes vs MCU V-frames.*

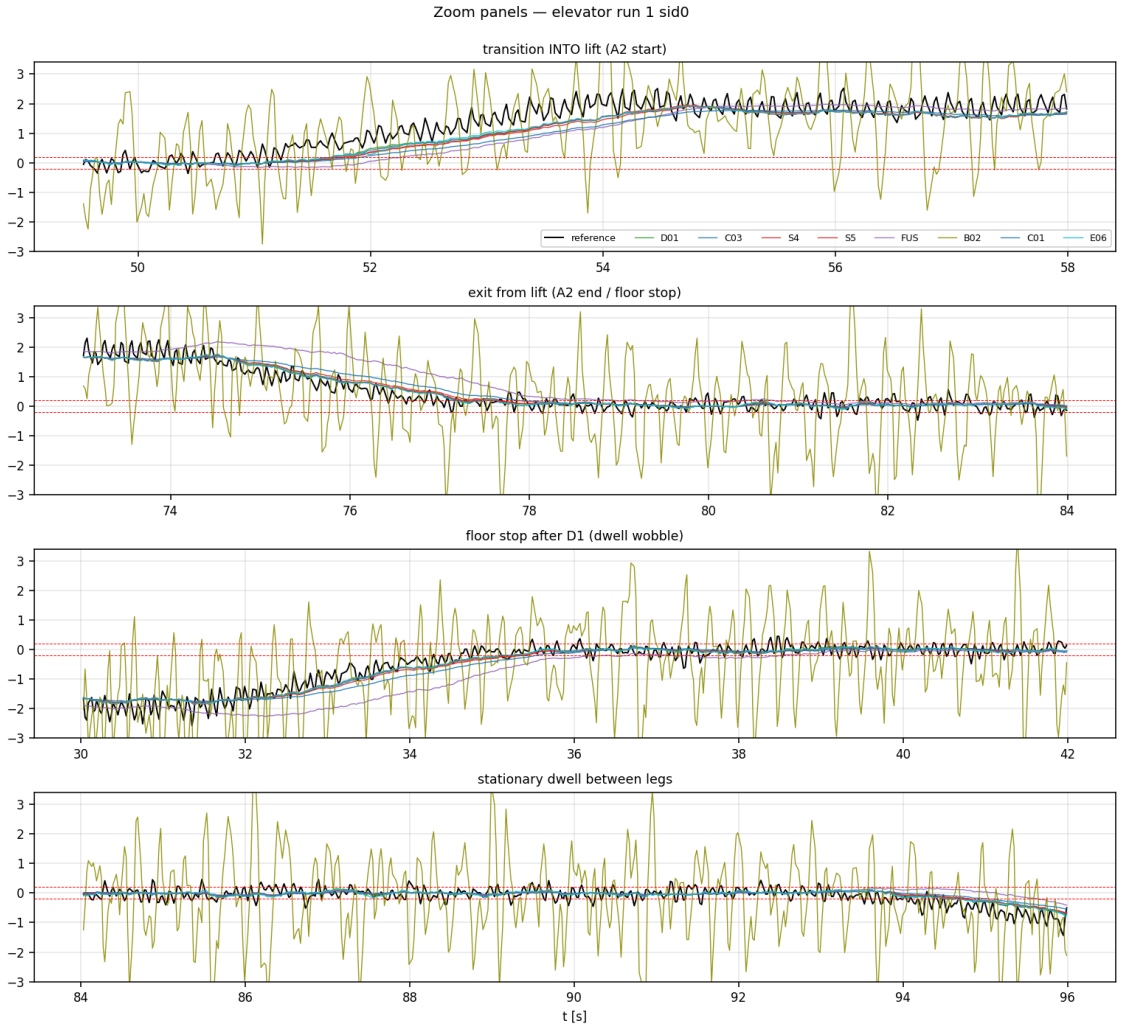



***Figure 3.*** *Zoom panels: transition into lift, exit/floor stop, and dwell behaviour for representative candidates.*

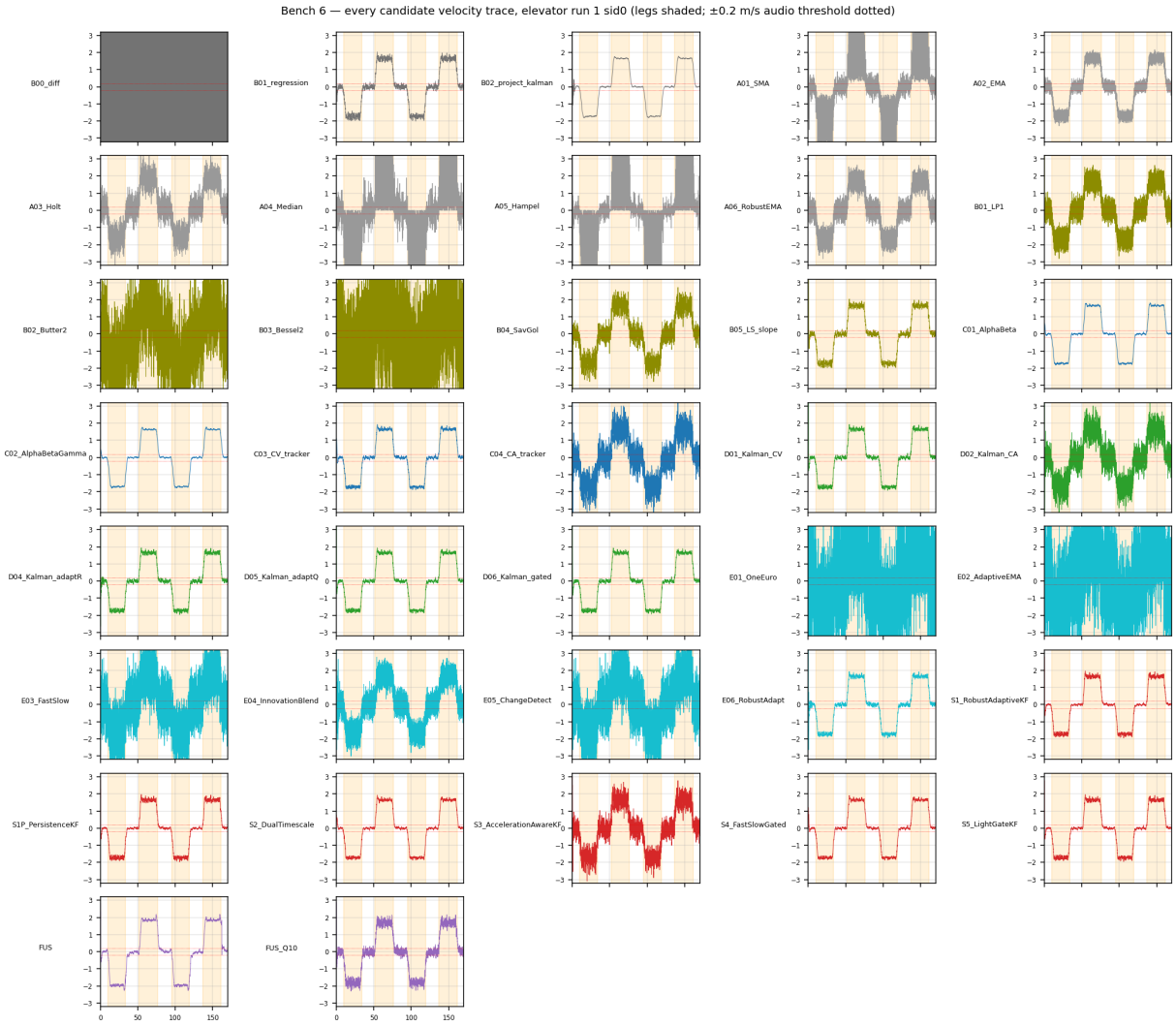



***Figure 4.*** *All 37 candidate velocity traces on run 1 (legs shaded; ±0.2 m/s audio thresholds dotted).*

## 3.3 Handtest, virtual audio, and CPU

Handtest: fastest onsets S2/S4/D01 (0.07–0.08 s); slowest S1/S5/FUS (0.21–0.23 s); C03/C01 unscored (event-metric artefact). Slow sustained blocks (66–69, 108–114 s) tracked with |bias| ≤ 0.05 m/s by the bank. Toss recovery: all bank filters re-settle within the metric window. Static-silence false latches (relative ranking): S2 18/h, S1 166/h, bank ≈310–350/h, FUS 629/h.

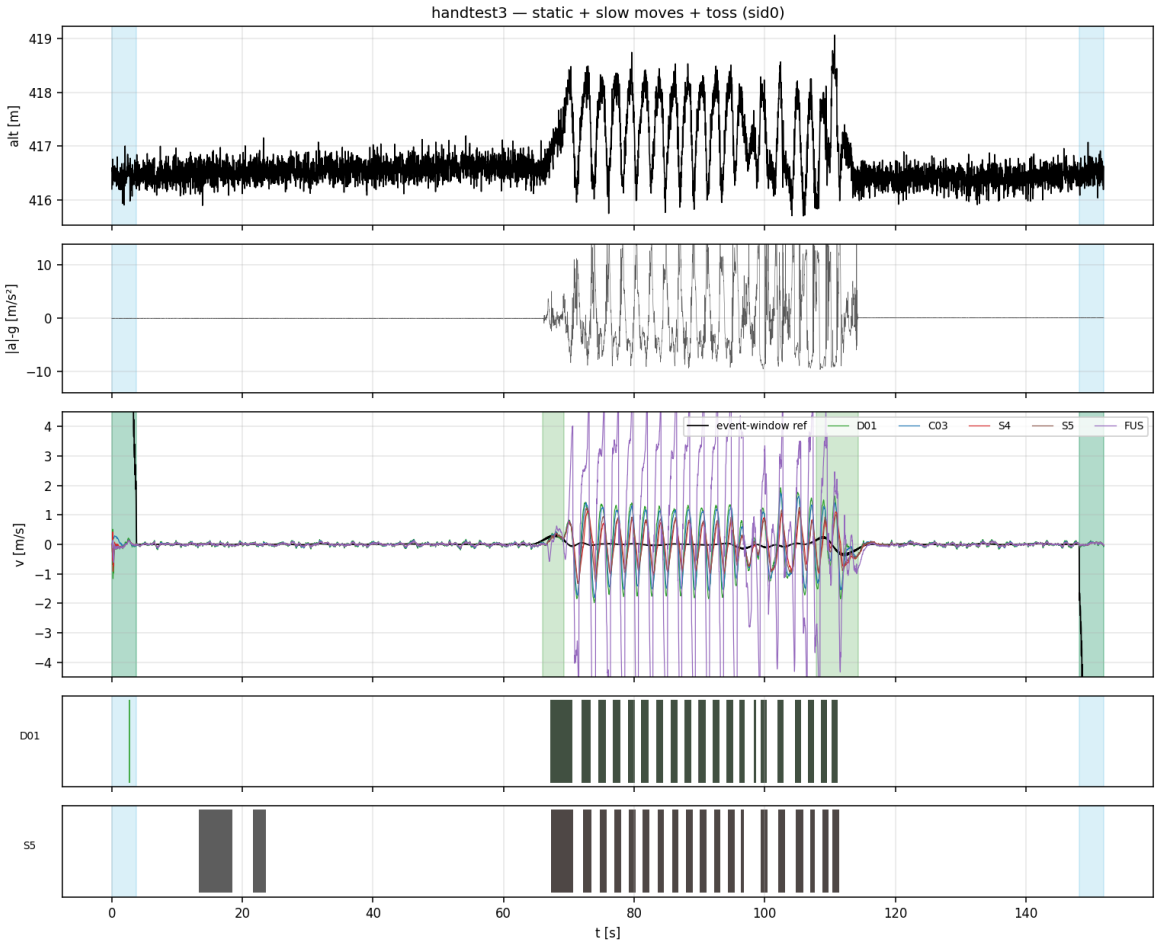



***Figure 5.*** *Handtest3: altitude, IMU acceleration, bank velocity traces with event and slow-block shading, and virtual-audio overlays.*

Virtual audio, separately: dwell false-ON between legs 2.0–2.4 s/run across the viable family (FUS 4.95) — the post-p0fix residual of the DEF-001/007 mechanism (Phase-era ≈16.5 s/run on the defective reference). Replay-vs-MCU transition agreement (frozen era, ±30 ms window): D01 84 %, S4 43–75 %, S5 33–62 %, dominated by timing near the asymmetric tone clamp rather than state disagreement.

***Table 4.*** *MCU CPU from B-line telemetry during the clean captures (real pipeline, no synthetic timing). Pre-p0fix lines carry the C03 refit storm (~1329/2671 µs, DEF-002 historical evidence). Efficiency is not a differentiator.*

| Estimator (on-target, post-p0fix) | sid0 avg/max µs | sid1 avg/max µs | tick budget |
|---|---|---|---|
| D01_Kalman_CV | 36 / 44 | 27 / 37 | 0.7 % |
| FUS (baro side; DEF-009) | 33 / 37 | — | 0.7 % |
| S5_LightGateKF | 77 / 80 | 59 / 88 | 1.6 % |
| S4_FastSlowGated | 97 / 101 | 69 / 82 | 2.0 % |
| C03_CV_tracker | 105 / 132 | 78 / 102 | 2.6 % |

# 4. Sampling-rate control

Decimating the clean streams to ≈25/20 Hz and 10 Hz: seconds-parameterized filters (D01, C03, S4, S5, FUS) are near-invariant (leg RMS 0.09→0.13–0.14 at 10 Hz, rise unchanged, exit inside the dwell-wobble floor). The fixed-coefficient B02_Butter2 does not transfer (leg RMS 1.49 m/s at 50 Hz vs 0.32 at 10 Hz) — its Phase 2–5 coefficients are wrong for the ESP-IDF rate; it is excluded from cross-chain comparisons. The ESP-IDF vs Arduino rate differences are therefore a non-issue for the Kalman/tracker/stacked families.

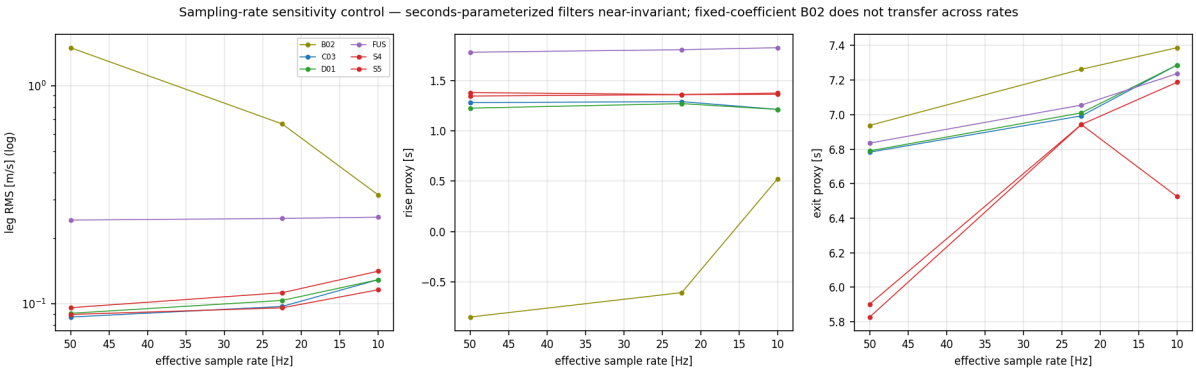



***Figure 6.*** *Rate sensitivity: seconds-parameterized filters near-invariant; fixed-coefficient B02 does not transfer across rates.*

# 5. Leakage-controlled tuning (NOT a freeze)

***Table 5.*** *Selection on train only (run1 + handtest3); every configuration reported on validation (run2 + statics). The train picks transfer — no leakage signature.*

| Family | Train pick | Validation effect vs frozen |
|---|---|---|
| D01_Kalman_CV | q=0.3 | rest RMS 0.057→0.036, exit 6.8→4.8 s, rise 1.19→1.48 s |
| S5_LightGateKF | q=0.1 | rest RMS 0.037→0.025, exit 5.7→3.7 s, rise 1.33→1.65 s |
| S4_FastSlowGated | tau_fast=0.5 | mixed (exit better, rise/rms worse) — not recommended |
| C03_CV_tracker | frozen best | all alternatives ≥ +18 % composite |

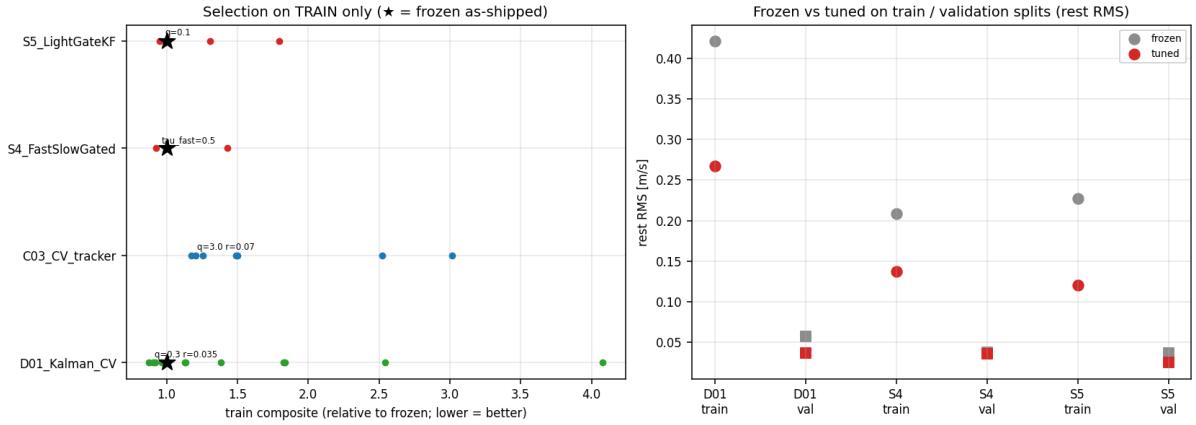



***Figure 7.*** *Tuning pass: train-side selection and train/validation rest-RMS for the frozen vs tuned configurations.*

# 6. Rankings vs Phases 2–5 and failure modes

Composite (7 metrics, percentile-normalized): **C03, C01, S2, S1, S4** top-5; D01 10th, S5 11th, FUS 19th. Per-metric leaders: rest noise B02_project_kalman / C02 / C01; response E06; exit C02/C01 (floored proxy); robustness cluster S1/S4/S5 within noise of each other.

What changed and why: (1) D01's rest floor 0.101→0.057 m/s — the old number was the defective reference, not the filter. (2) S5's Phase-4 stand-out status does not survive corrected streams: the S1/S4/S5 cluster is tight (0.035–0.037) and S5's onset is slow. (3) S1's missed-lift weakness is gone on these captures but its onset is now slowest. (4) C03 rises to the composite top — its Phase-2/3 weaknesses were firmware-era artefacts (DEF-002 CPU, DEF-001 false-lift), both fixed. (5) FUS shows no measurable benefit on vibration-free data; the fusion value proposition stays open for real-flight data. (6) B02's collapse is a rate artefact.

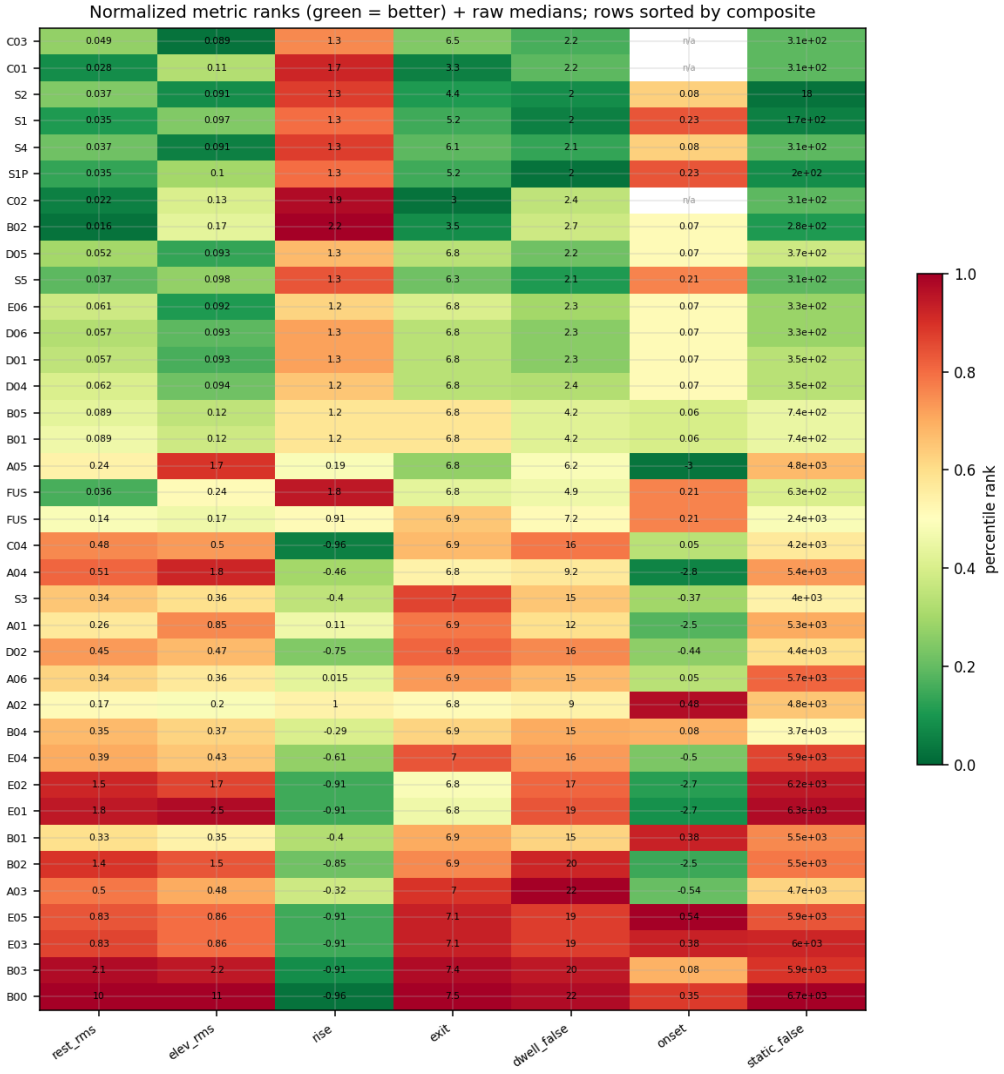



***Figure 8.*** *Normalized metric ranks (rows sorted by composite) with raw medians.*

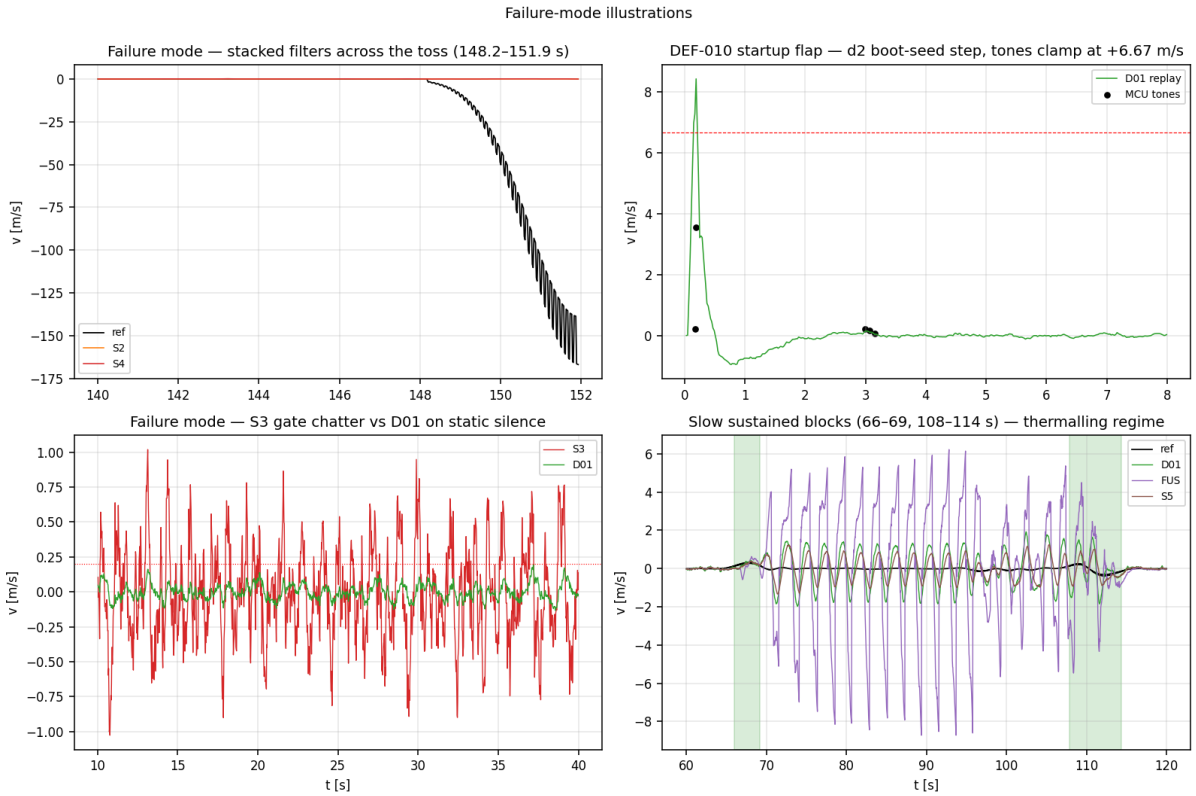



***Figure 9.*** *Failure modes: S2 freeze-trap across the toss, startup flap (DEF-010), S3 gate chatter, slow-block behaviour.*

# 7. Limitations and next phase

No subsecond ground truth on the elevator: rise/exit are labelled proxies [D]; exit is floored at ≈6.8 s by dwell wobble. Handtest static window contains operator micro-movements; false-latch rates are relative rankings. Post-p0fix dynamic captures do not exist yet — S4/S5/FUS MCU audio equivalence on transitions is vacuous, not failed. Onset NaN (C03/C01) is an event-metric artefact, reported as unscored. Fusion evaluation is baro-reference-bound. The two clones differ; all medians are reported over both sids.

**Next phase (BMP581 + LSM6DSV16X):** register the new captures in datasets.py and re-run the pipeline; the registry is the only sensor-specific file. The reference firmware for the current sensor set is archived at esp-idf/firmware_archive/vario_idf_MS5611x2_MPU6050_p0fix_r8e79e68_20261023/ (binaries + SHA-256 + README) for A/B flashing. DEF-011 must be fixed in the sensor-integration build before any absolute-altitude comparison.